

Experiment No : 10

Experiment Name : Study of astable multi-vibrator using 555-timer

Objectives:

- To implement the astable multivibrator using 555 timer
- To observe the output waveforms of the astable multivibrator

Theory:

In astable mode, the 555 timer operates as a free running oscillator that continuously switches between high and low states, generating a square wave output. This makes it useful for applications such as clock pulses, LED flashers and frequency generators. The output frequency is given by,

$$f = \frac{1.45}{(R_A + 2R_B)C}$$

Required Apparatus:

- 555-timer (NE-555) - 1 piece
- Resistors (10 K Ω - 2 piece)
- Capacitor (1 μ F, 10 μ F - 1 piece each)
- Oscilloscope
- DC power supply (0-15V)
- Project board
- Multimeter
- Connecting wires

Circuit Diagram:

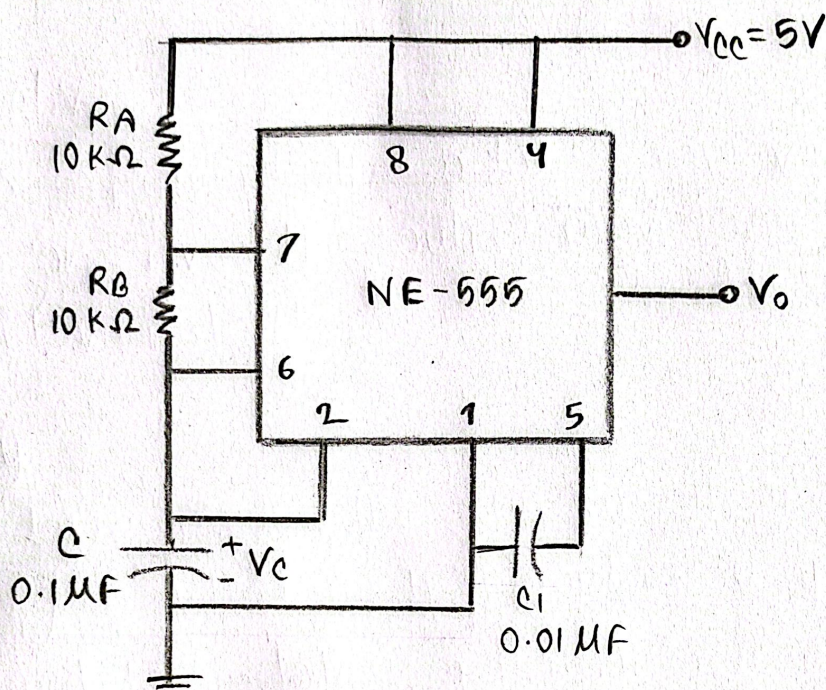


Fig: Circuit Diagram for implementing astable multi-vibrator circuit using 555-timer

Procedure:

- (i) The circuit was connected according to the figure
- (ii) Necessary power supply was given and the corresponding output waveform was recorded
- (iii) The obtained frequency of oscilloscope was compared with the theoretical frequency

Calculations:

Theoretically, $f = \frac{1.45}{(R_A + 2R_B)C} = 483.33 \text{ Hz}$

Experimentally, $f' = \frac{1}{t} = \frac{1}{2.16 \times 10^{-3}} = 462.96 \text{ Hz}$

$$\therefore \text{Error} = \frac{1483.33 - 1462.961}{1483.33} \times 100 \%$$
$$= 4.21 \%$$

Discussion and Conclusion:

In this experiment, the 555-timer was successfully implemented and it operated as expected. The small error (4.21%) confirms good arrangement between theory and experiment.

Precautions:

- All the connections should be correct
- 555 timer terminals must be identified properly
- Calibration must be performed properly